



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CS3501 Compiler Design

Name of the Faculty member (s): Dr.I.Gethzi Ahila Poornima , AP(SG)/CSE

Innovative Practice Description

- **Unit / Topic:** Unit III / Basic Blocks and Flow graph

- **Course Outcome:** CO4

- **Activity Chosen:** Reflection

- **Justification:**

In order to perform or implement various optimization techniques on any intermediate representation of a high level language source code, partition of basic blocks and construction of flow graph is most required. As a teacher, I should ensure that all the students are clear with this concept.

- **Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**

- At the conclusion of the session, this activity was carried out in a classroom for total of 15 minutes.
- The students learned about "Partitioning basic blocks by selecting leaders and constructing flow graph" with the help of sample code.
- The students were instructed to write down any questions or level of understanding the concept.
- The questions or the topics that were not understood were written by them and submitted. It is shown in Fig. 1.
- After reading the responses, the student's queries were clarified.
- A number of students had questions, and those questions were clarified in class the following day for everyone.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO12	PSO1
CO3	3	1	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PSO1
	3	1	1	1

Justification for correlation	The idea of designing basic block and constructing flow graph enables the students to solve various problems related to code optimization.	The students examine the flow graph and can analyze the liveliness of variables in the program.	The students find the solutions for finding live variable analysis and local and global sub expression elimination.	The Student's concise writing of their questions helped them develop their skills.	The students apply the construction of flow graph in further code optimization techniques.
--------------------------------------	--	---	---	--	--

• **Images / Screenshot of the practice:**

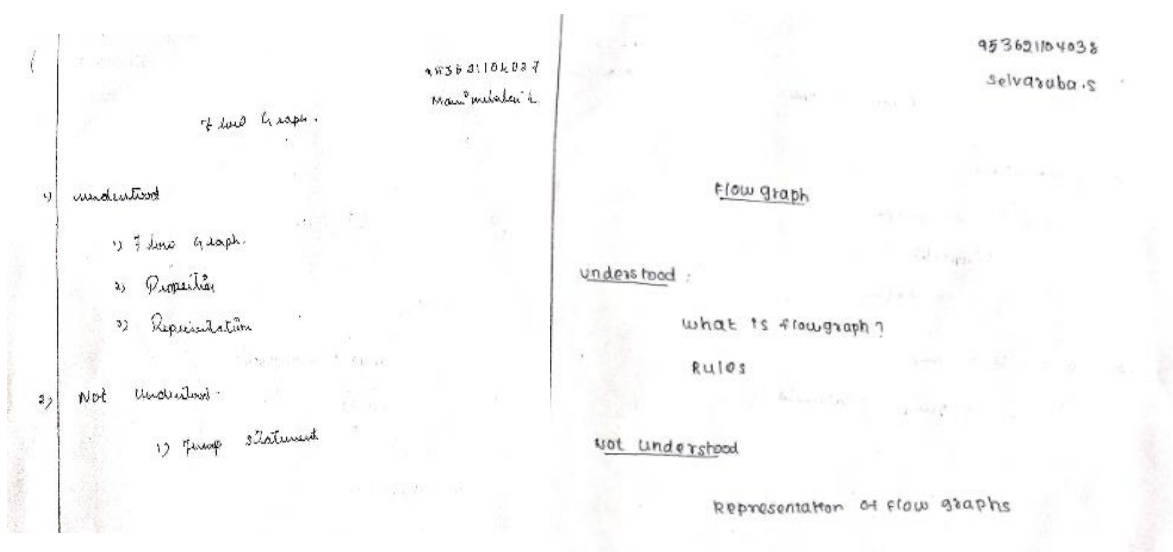


Fig 1: Reflection activity

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- The students were allowed to evaluate themselves and identify areas where they needed further clarification.
- It was simple to identify the subtopic that many students found confusing.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- This activity makes it simple to identify students who didn't get the fundamentals of constructing basic blocks for the given sample code and to provide them with further explanations in the subsequent topics (Live variable analysis, local and global optimization) lesson, so raising their comprehension level.

❖ **Challenges faced in implementation:**

- I wasn't able to get a true picture of all the students' viewpoints because some students opted not to ask any questions.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/4_OIT551_Reflection.pdf
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/Reflection-HCI-MSS.pdf>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2021-2022/BV_CS8602_Reflection.pdf

Signature of Faculty Member**HOD**